

Native MXFP4 Dataflow for Persistent Gated DeltaNet Decode on a FPGA

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Abstract—Recently, Large Language Models (LLMs) face significant limitations due to both massive computational requirements and memory scalability. The Transformer model’s Key-Value (KV) cache is directly proportional to the input sequence length, which causes serious constraints when using available hardware. Hybrid architectures reduce reliance on the full-attention KV cache by replacing some layers with fixed-state linear attention; however, using high-precision data types (e.g., FP32) quickly exhausts on-chip memory and requires repeated data transfer from off-chip memory, resulting in a significant number of expensive data transfer operations and potentially large amounts of wasted energy. Therefore, we propose a native 4-bit OCP Microscaling (MXFP4) datapath that can be used to reduce memory usage through compression of fixed-size linear state matrices. Our proposed MXFP4 datapath also minimizes the amount of time spent waiting for memory access by maximizing on-chip data retention. We evaluated our proposed MXFP4 design on a Xilinx Alveo U55C FPGA. Results from our evaluation show that our MXFP4 datapath achieves a 97.4% reduction in power consumption relative to an H100 baseline, and an 86.7% increase in memory efficiency relative to FP32, thereby enabling us to accelerate hybrid attention mechanisms while maintaining acceptable synthetic numerical similarity.

Index Terms—Large Language Models, Hardware Accelerator, FPGA, MXFP4, Linear Attention, Hybrid Transformers, Memory Hierarchy Partitioning, Xilinx Alveo U55C.

I. INTRODUCTION

The response of an AI agent can take up to seconds before finally delivering an answer. To understand the reason for this kind of delay, we must first realize the framework being used at the front of modern LLM operations. Many of the most popular models, such as ChatGPT and Claude, were founded through the transformer model architecture [6], a completely software-based mathematical technique that processed data sequences simultaneously using a “Self-Attention” mechanism that decided how much one word would relate to another to form context and reference. When a new word is generated,

the model goes to look back on previous words, or “tokens”. This was sorted out by saving all that information into a mathematical representation into external memory which would be called a key-value cache (KV cache) [2].

Naturally, when the conversation becomes longer, then the KV would also be bigger which would slow down the processing time of the model [7], thus the development of another mathematical alternative called the linear attention, like the Gated DeltaNet (GDN) model [8]. This linear-time sequence model compresses the history of data down into a single fixed grid known as a recurrent state matrix S [3]. The tradeoff with the linear-attention layer is that the very specific details of the past history gets simplified a lot. However, by combining both approaches and stacking them, researchers have made the token go through linear layers first then go into a full-attention layer, known as the 3:1 structure block, to more efficiently process previous history while maintaining data integrity [9]. These two are defined with an algorithm complexity with respect to sequence length, N , as $O(N)$ for a full-attention layer and $O(1)$ for a linear-attention layer for one token [8]. That means the number of MAC operations for large scale token history would be lower since it is a constant when compared to the growing operation count of the full attention layer. It should be noted that the total algorithm complexity on an entire execution sequence has the full attention layer scale quadratically $O(N^2)$ and the linear attention layer scale to $O(N)$. To keep accuracy while being efficient, the present industry’s choice to use the previously mentioned 3:1 structure doesn’t eliminate the nature of the growing complexity, but rather reduces the growth of complexity as compared to a scenario of pure full attention layers [9].

This specific replacement technique targeting the growing KV caches of each internal macro block of a block layer structure does make the memory usage a lot more efficient than without [10]. However, traditional implementations utilizing high-precision data formats such as FP32 will quickly saturate the limited memory on these kinds of FPGAs [4]. When

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these state matrices start to exceed the local capacity, the hardware must go on to use external bandwidth memory. To break this memory wall and improve efficiency of current integer standards, we implement a 4-bit MXFP4 datapath onto a Xilinx Alveo U55C accelerator to create an efficient connection between the local memory and external memory [5]. This can be done by treating the URAM and BRAM as the permanent execution area for those compressed S matrices and reserving the high bandwidth memory (HBM) for the growing KV caches of the full attention layers.

By maximizing the efficiency of the data format itself, we can make a better case for the limited on-chip storage available. Numerical states can be packed into extremely compact 4-bit OCP microscaling formats through the specific utilization of 4-bit E2M1 elements that are bounded by a shared 8-bit E8M0 block scale [11]. The processing elements will be able to perform continuous linear attention updates without having to request a time-consuming demand to the external memory bus. By focusing on one piece of a hybrid attention model, we can improve the performance and efficiency of the entire implementation that is being used. In a hybrid model like the previously mentioned 3:1 structure, we can look at the primary building block that makes up 75% of the core pipeline and focus entirely on improving what makes up the bulk of the model which naturally means optimizing the entire system as a whole [9].

To recap the main discussions made before, the primary objective of this research is to maximize the use of low-latency on-chip memory resources that maintains the precision needed for long context models. By mapping 4-bit MXFP4 multipliers directly onto the LUT resources of the FPGAs, parallel computation is achieved [12]. This paper effectively introduces a new data packaging method optimization that can break the modern memory wall and unlock higher potential of hybrid linear attention architectures on high-end FPGA fabric [5].

II. HARDWARE ARCHITECTURE

A. Decoding Layer Overview

One Gated DeltaNet decoding layer is implemented for sequential autoregressive inference. In addition, for each decoding step, the kernel takes query, key, value, beta, and gate inputs and updates the recurrent state. Finally, it outputs the vectors.

In contrast, the majority of architectures keep the recurrent state in FPGA memory throughout all token steps, avoiding multiple movements of the entire state matrix through external memory [10]. Therefore, the state-updating operation is done as a fixed state-updating pipeline.

B. Persistent Recurrent State

As opposed to an existing KV cache that expands in size as sequence length increases, the recurrent state of Gated DeltaNet consists of a fixed-size recurrent-state matrix [3], [8]. The recurrent-state matrix stores the compressed information

of previous tokens, functioning as a compact associative memory that addresses the standard LLM memory wall [1], [4], [7]. The recurrent-state is updated after every decode cycle.

The recurrent state S_t is stored within partitions of on-chip FPGA memory banks [10]. With respect to the stored recurrent state S_t , it allows access to different parts of the state matrix in parallel. On-chip storage of the recurrent state results in reduced latency and energy consumption since each decode step will need to load the prior state; modify it; and save the modified version. However, accessing the full state matrix repeatedly off-chip and back to on-chip will become a resource-consumptive task, worsening the memory-bound limits of standard decode paths [7].

C. Gated DeltaNet Update Path

Every decode step uses the current key and value to update the recurrent-state. First, the prior state is multiplied by the key vector to create a prediction based on delta-rule dynamics [3]:

$$r_t = S_{t-1}k_t \quad (1)$$

The prediction is what the prior state already holds for the upcoming key. The hardware performs a comparison between the prediction and the new value vector:

$$e_t = r_t - v_t \quad (2)$$

The error (e_t) is created due to a difference between what was stored by the prior state and what information is presented by the new token. This error is scaled by the beta input and is used to update the state:

$$S_t = S_{t-1} - \beta_t e_t k_t^T \quad (3)$$

This state update is a rank-one update. It changes the state by applying both the direction of the key and the error against the value. Once the state is updated, a query vector is applied to produce the output:

$$o_t = S_t q_t \quad (4)$$

Thus, there are repeated applications of matrix-vector operations on S_t ; therefore, the updating of the recurrent-state and producing the output are both related to the same problem.

D. Five-Step Token Pipeline

Gated DeltaNet could naively be implemented over three steps. First, an old-state S_{t-1} would have been loaded so that a prediction could be made regarding how to retrieve data using a key vector. Next, the retrieved data would have been used to correct/update the state. Lastly, an updated-state would have been loaded so that an output could be produced using a query vector. The resulting design would be inefficient for an FPGA implementation because the largest structure (the state-matrix) must be traversed completely by every token.

However, we were able to optimize the sequence of calculations such that only partial traversal of the full state-matrix occurs, implementing a fused dataflow framework tailored for linear attention bottlenecks [10]. We did this by limiting use

of reusable terms derived from state-data used in computing retrieval terms to those computations used in generating an output; therefore, no additional full read of an updated-state needs occur for output generation purposes. Every token goes through five states.

First, the input preparation module puts each token in its corresponding scaled MXFP4-block. Second, the state-read module loads up the current-state S_{t-1} from parallel memory-banks in order to calculate retrieval terms for the delta-rule. Third, the delta-update module calculates beta-scaled error and state-corrective terms. Fourth, the write-back phase writes updated-state S_t directly back to memory-banks where they reside. Fifth, the output-generation module generates final-token-output o_t by adding query-vectors with previously-generated-state-derived-terms. As such, we eliminate a second-full read of-state.

Overall, our optimization reduces total-state-traversal work from approximately three full passes to less than two by collapsing these passes into a tighter, five-phase pipelined datapath design [10].

E. Native MXFP4 Datapath

We chose MXFP4 as our arithmetic datapath in order to reduce computation requirements and mitigate the memory wall [5]. MXFP4 represents each individual element as a four-bit E2M1 value and assigns a single shared E8M0-scale value to a designated block of elements. A block of MXFP4-values may be represented as follows:

$$x_i \approx 2^{s_b} \hat{x}_i \quad (5)$$

Here, s_b is the shared-block-exponent while $\hat{x}_i$ is a four-bit E2M1-element. Thus, MXFP4 provides greater range than standard INT4 while still storing each individual element at only four bits; thus making MXFP4 suitable for low-area FPGAs while minimizing resource usage and sustaining a functional numeric range [12].

F. LUT-Based E2M1 Multiplication

LUT-based E2M1 multipliers are utilized in the FPGA LUT logic as they avoid conversion of operands from BF16 (or FP32) format and provide a direct path to E2M1 multiplications. Each E2M1 is split into three parts: Sign, Exponent, and Mantissa. An XOR operation is performed on the Sign field. A biased sum of Exponents is calculated. The Mantissas are multiplied together and then normalized. The resulting fixed point partial-product is prepared for accumulation. Additionally, this process minimizes the physical area of the inner Multiplier Block and eliminates the requirement for large floating-point DSP Blocks that would be necessary to process four-bit inputs.

G. Block-Exponent Alignment

Since MXFP4 employs shared block-scales, the magnitudes of the products of two blocks can vary. After generation of partial-products, these values are therefore adjusted with

regard to their block-exponents prior to summation. If two blocks have exponents s_a and s_b , then

$$s_{prod} = s_a + s_b \quad (6)$$

is the combined exponent of the product. Next, the partial-product will be shifted-left so that it is placed in the correct range for summation with an accumulator. This approach ensures the numeric correctness of the scaled values in MXFP4.

H. Design Parameters

Three design parameters determine the behavior of the overall design architecture: Block Size (B), Key Parallelism (P_K) and Value Parallelism (P_V).

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REFERENCES

- [1] [Placeholder: Papers on the LLM “Memory Wall” and latency limits (e.g., Pope et al. / Sheng et al.)]
- [2] [Placeholder: Foundational Transformer / KV Cache bounds (e.g., Vaswani et al.)]
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- [8] [Placeholder: Gated DeltaNet definitions]
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